2017 Ph H2 Q12

Section: Particles and Waves

Topic: Interference

A student carries out an experiment using **two coherent loudspeakers** to demonstrate **interference of sound**.

(a)

Explain what is meant by coherent sources.

✅ **Coherent sources** are sources that emit waves with a **constant phase relationship** and the **same frequency**.

> Coherence allows **stable interference patterns** to form — crucial for producing constructive and destructive interference.

(b)

Describe how the student could detect a point of destructive interference.

✅ The student walks in front of the speakers and listens for a point where the **sound becomes very quiet** or **disappears** — a **minimum** in sound intensity.

> At this point, waves from both speakers arrive **out of phase** and **cancel** each other out.

(c)

Explain in terms of path difference why destructive interference occurs at this point.

✅ Destructive interference occurs when the **path difference** between the two waves is:

$$\left(n + \frac{1}{2}\right)\lambda \quad \text{where } n = 0, 1, 2,$$

This means the waves arrive **180° out of phase** and interfere destructively.

🧠 Revision Tips:

- **Destructive interference** happens when crest meets trough → cancellation
- **Path difference** is the difference in distance travelled from each source
- At a **minimum**, path difference = $\frac{\lambda}{2}, \frac{3\lambda}{2}, \ldots$
